

Competitive Programming



Detailed
Course Syllabus

WEEK 01:

- Intro to CP, time complexity, basic concepts, Searching, Sorting, CP Basic techniques, etc.
- Prefix Sum, Binary Search, Euler Totient Function.

WEEK 02:

- Primes, Binary Exponentiation, LDE, etc.
- Sieve, Segmented Sieve, Wilson's Theorem.
- Modular Arithmetic, Combinatorics.
- Bit Manipulation, Shortcuts used in CP.

WEEK 03:

- Recursion.
- Backtracking(Basic to Advanced).

WEEK 04:

- Stack, Queue, Priority Queue.
- Divide and Conquer and its related algorithms.

WEEK 05:

- Greedy Algorithms.
- String Algorithms.

WEEK 06:

- Trees (Covering all algorithms and tricks related).
- Breadth first search.
- Depth first search.

WEEK 07:

- Graphs (Covering all algorithms and tricks related).

WEEK 08:

- Dynamic Programming-I (Basic Medium DP).
- Dynamic Programming-II (Hard DP - Probability DP, Digit DP, Bitmask DP).

WEEK 09:

- Disjoint Set and Sparse Table.
- Fenwick Trees/BIT.
- Segment Tree (Complete).

WEEK 10:

- LCA.
- Sqrt Decomposition.
- Mo's Algorithm